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2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence [SWP: *The potential*
4 *impact of personalized AI assistants on collective*
5 *intelligence?*]

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11 Abstract

12 Recent advances in artificial intelligence (AI) have enabled widespread use of per-
13 sonal AI assistants for various cognitive tasks, ranging from simple everyday tasks
14 to scientific discovery and creative writing. While AI assistance can improve the
15 performance of each individual, it can weaken their subsequent performance with-
16 out AI and further limit the diversity of interests and opinions. These risks raise
17 the question of whether human collective intelligence can persist as our society be-
18 comes increasingly reliant on AI assistants. To address this question, we develop an
19 evolutionary model of collective intelligence, where each individual participate in a
20 prediction task, consult AI, and imitate each other’s prediction strategy. We find
21 that “Chatbot” AI, which depends on users’ interests, can prevent the emergence
22 of collective intelligence by either limiting diversity or propagating systematic bias
23 through overreliance. Directly incentivizing or penalizing AI use generally worsens
24 these outcomes. Instead, a balanced incentive structure that preserve minority ex-
25 pertise and reward contributions to collective outcome can promote the emergence of
26 collective intelligence. However, the inherent ability of the population’s to make pre-
27 diction in the absence of AI remains suboptimal even under this balanced incentive
28 structure, raising concerns for continued AI reliance. [SWP: *conclusion?*]

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual participates in a prediction task from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

70 Mathematical model

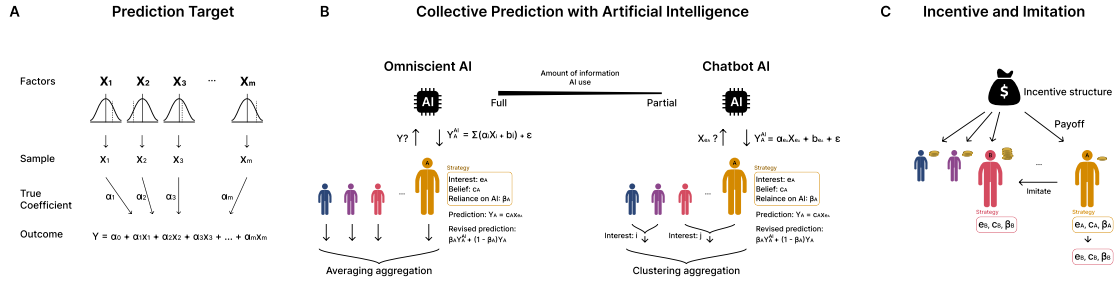


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

We begin by extending previously established models of CI, in which a group of players participates in a prediction task (Figure 1): each player makes an individual prediction based on partial observations of the underlying causal factors, and the group forms a collective prediction by aggregating these individual predictions [30, 12, 13]. Specifically, the prediction target, Y , is modeled as a linear combination of M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution (Figure 1A). Then, each individual A observes a single factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then independently consult their AI assistants and revise their prediction based on their reliance on AI, β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are then aggregated using two alternative aggregation rules, averaging and clustering, to form collective prediction.

To model AI, we assume that AI has already been trained on similar tasks and is able to make predictions based on partial or full observations of causal factors (Figure 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on its internal parameters, which can include factor-specific bias (b_i) and stochastic error ϵ . Here, we consider two extreme types of AI response, which differ in the amount of information AI uses when making predictions. In one extreme, the AI can observe all causal factors and use all information to make predictions, regardless of the user's interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works like a chatbot, where the AI predicts the outcome based only on the user's interest:

93 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 94 and ‘Chatbot AI’, respectively.

95 We assume that appropriate aggregation rule for collective prediction depends
 96 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 97 single factor. Thus, we use clustering aggregation, under which revised predictions
 98 are first averaged among individuals who are interested in the same factor, and
 99 then summed across factors. When Omniscient AI is used, however, each revised
 100 prediction contains information about all causal factors, in which case clustering
 101 these predictions by interest would be inappropriate. Instead, we use averaging
 102 aggregation, under which revised individual predictions are averaged directly to form
 103 the collective prediction.

104 Each generation consists of multiple round of predictions, and each player is re-
 105 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 106 we assume that a sufficiently large number of prediction rounds occurs within each
 107 generation, allowing us to use the expected payoff value rather than computing the
 108 payoffs for every round. We consider two incentive structures from previous work,
 109 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 110 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 111 to individuals who can push the collective prediction toward the true target, regard-
 112 less of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In contrast, the
 113 niche-expert structure gives higher payoff to individuals who have both better pre-
 114 dictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A} denotes
 115 the proportion of players interested in the same factor, thereby allowing the niche-
 116 expert structure to preserve interest diversity. We note collective intelligence fails to
 117 emerge under the combination of averaging aggregation with niche-expert structure
 118 in the absence of AI [13].

119 Through time, the group evolves by allowing each individual to imitate peers
 120 who received higher payoffs (Figure 1C). Specifically, after each generation, two
 121 individuals are randomly sampled, and one imitates all three strategies (the factor of
 122 interest, factor belief, and reliance on AI) of the other with a probability determined
 123 by the difference in their expected payoffs (Materials and Methods). Individual
 124 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 125 rules. Collective accuracy is defined as the reduction in prediction error achieved by
 126 the collective prediction relative to a baseline prediction α_0 : $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective accuracy reaches a near
 127 perfect level.
 128

129 This model now allows us to explore the impact of AI on CI depending on the
 130 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 131 of CI with and without AI and subsequently investigate a how different incentive
 132 structures affect the emergence of CI in the presence of AI.

AI can either promote or prevent the emergence of CI

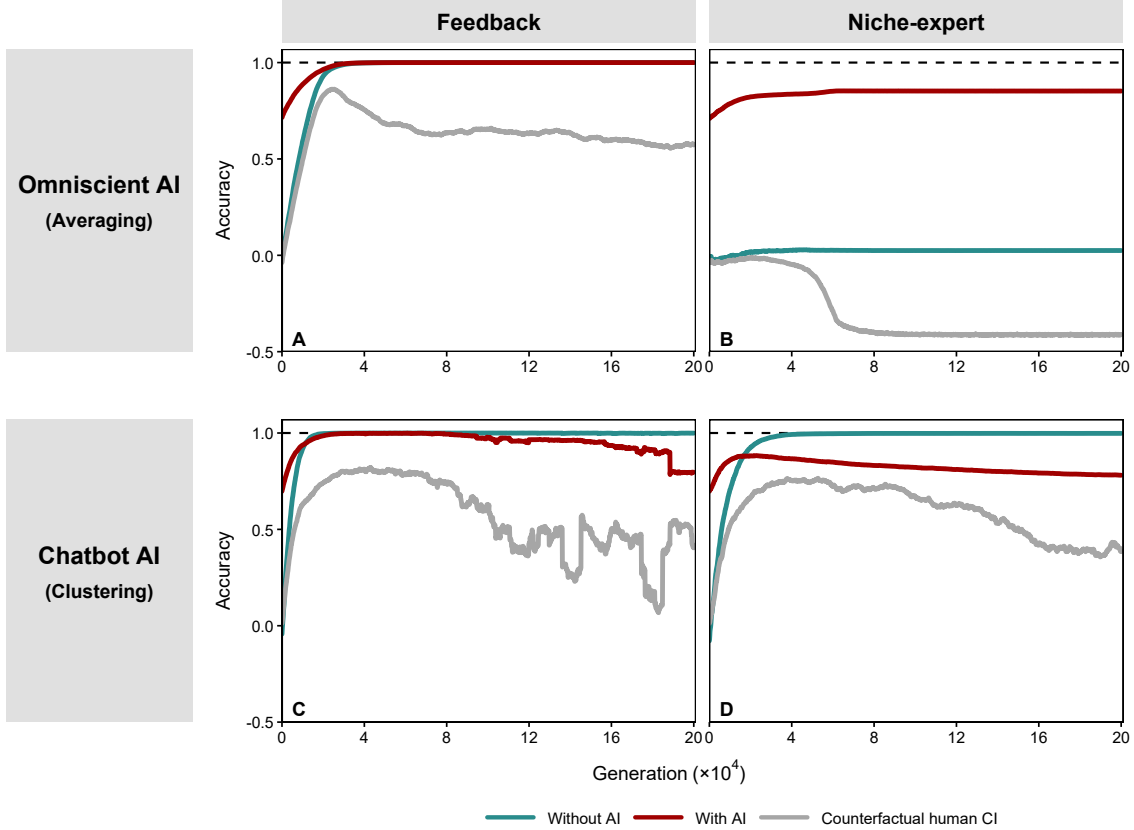


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

To understand the impact of AI, we first compare collective dynamics with and without AI. First, consistent with [13], simple imitation rules and incentive structures allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One exception is the use of simple averaging with niche-expert incentive structures, under

139 which the collective accuracy of the population remains near zero. [JS: This is
140 because enhancing individual accuracy causes individuals to predict close to the
141 true coefficient of their interest, but simply averaging these predictions does not
142 necessarily produce an accurate estimate of the full prediction target.] (Fig. 2B).

143 We find that AI usage can have heterogeneous outcomes depending on its type.
144 For example, the use of Omniscient AI can either preserve CI under the feedback
145 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
146 expert incentive structure (Fig. 2B). These results are robust across different levels
147 of AI accuracy and can be shown analytically by assuming a sufficiently large group
148 of players (Fig. S1; see SI Evolutionary dynamics section). [SWP: need to get
149 back.]

150 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
151 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
152 structure, the use of Chatbot AI leads to a loss of diversity in the population's
153 interest, making it impossible to account for effect of some causal factors (Fig. S1,
154 S2). This causes an increase in collective variance and stochastic decrease in collective
155 accuracy (Fig. 1; Fig. S1). We note that the loss of interest diversity may arise even
156 in the absence of AI in finite populations (Fig. S3), but the use of Chatbot AI
157 makes the group more vulnerable to such loss (Fig. S3, S4). Under the niche-
158 expert incentive structure, the population becomes overly reliant on AI, causing the
159 collective prediction to inherit the AI's bias (Fig. S2). This leads to an increase in
160 collective bias and a decrease in collective accuracy (Fig. 2; Fig. S2). Interestingly,
161 we find that the niche-expert structure can prevent the use of AI under specific
162 assumptions about the relationship between the true coefficient and AI bias (Fig.
163 S1). In this case, the population completely excludes AI assistance, allowing the
164 CI to emerge eventually. However, this phenomenon can substantially delay the
165 emergence of CI (Fig. S5).

166 To investigate whether the group can retain its predictive ability when AI assis-
167 tance suddenly becomes unavailable at any given time, we quantify counterfactual
168 collective accuracy based on unrevised predictions. Across all scenarios, we find that
169 the counterfactual collective accuracy is lower than the collective accuracy of the
170 corresponding population.

171 In summary, an omniscient AI can promote CI, but AI operating with limited in-
172 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
173 ability of the population to make accurate predictions on their own, raising concerns
174 about future AI usage. These findings further underscore the need for incentive
175 structures that can mitigate the potential risks associated with AI assistance.

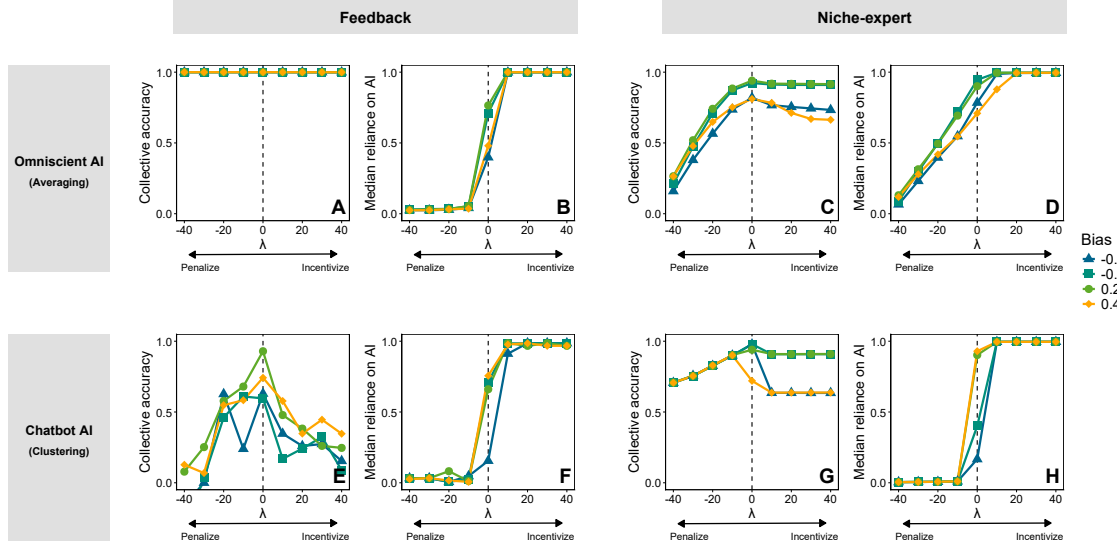


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

176 Directly incentivizing or penalizing AI use can ad- 177 ditionally harm CI

178 Given the strong influence of AI on collective outcomes, a straightforward inter-
179 vention is to directly incentivize or penalize the use of AI. To investigate whether
180 such direct incentives or penalties can help the emergence of CI, we modify the two
181 incentive structures by adding a payoff term proportional to individual reliance on
182 AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

183 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP:
184 maybe we need to call incentive pi earlier on...] [SWP: also good to define beta
185 again here as a reminder.]

186 Directly incentivizing or penalizing AI use can further reduce the collective ac-
187 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback
188 structure; in the latter case, the population retains collective intelligence regardless
189 of the intervention strength (Fig. 3A). The reduction in collective accuracy arises be-
190 cause incentivizing or penalizing AI usage causes individuals to imitate others based
191 on their AI usage rather than their predictive accuracy. Thus, stronger interventions
192 can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H)
193 but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When
194 AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but
195 this represents a narrow parameter regime (Figure 3G).

196 Overall, these results indicate that simply incentivizing or penalizing the AI, in-
197 dependent of individuals’ performance, use may exacerbate rather than resolve the
198 risk of AI on CI. Thus, we also considered an incentive structure where interven-
199 tions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to
200 individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

201 **Balanced incentives mitigate AI-induced failures of** 202 **CI**

203 Neither the original incentive structures nor simple interventions targeting AI use are
204 sufficient to promote CI. What incentive structure, then, would work in the presence
205 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
206 the feedback and niche expert incentives.

207 The core idea behind the balanced structure is to maintain interest diversity and
208 prevent collective bias. On one hand, the feedback incentive structure is prone to
209 the loss of interest diversity but can prevent collective bias by incentivizing players
210 whose predictions lie opposite to the collective prediction. On the other hand, the
211 niche-expert structure can maintain interest diversity because it directly incentivizes
212 minority interests. However the niche-expert structure cannot prevent collective bias
213 because, within the same factor of interest, individuals are rewarded only according
214 to their individual predictive performance, which in turn homogenize the predictions.

215 Therefore, when the two incentive structures are combined, the population achieves
216 CI by maintaining interest diversity while limiting collective bias (Fig. 4A; Fig. S3).
217 Under the balanced structure, individuals remain more evenly distributed across
218 causal factors, and their beliefs lie closer to the true coefficients than under either
219 incentive structure alone (Fig. 4C–E). The balanced structure also produces higher
220 counterfactual collective accuracy than the feedback or niche-expert structure (Fig.
221 4B). These patterns are driven by the emergent property of the balanced structure,
222 under which the population increases its reliance on AI as AI becomes more accurate.

223 [SWP: *may need some robustness check rather than a case study...*]

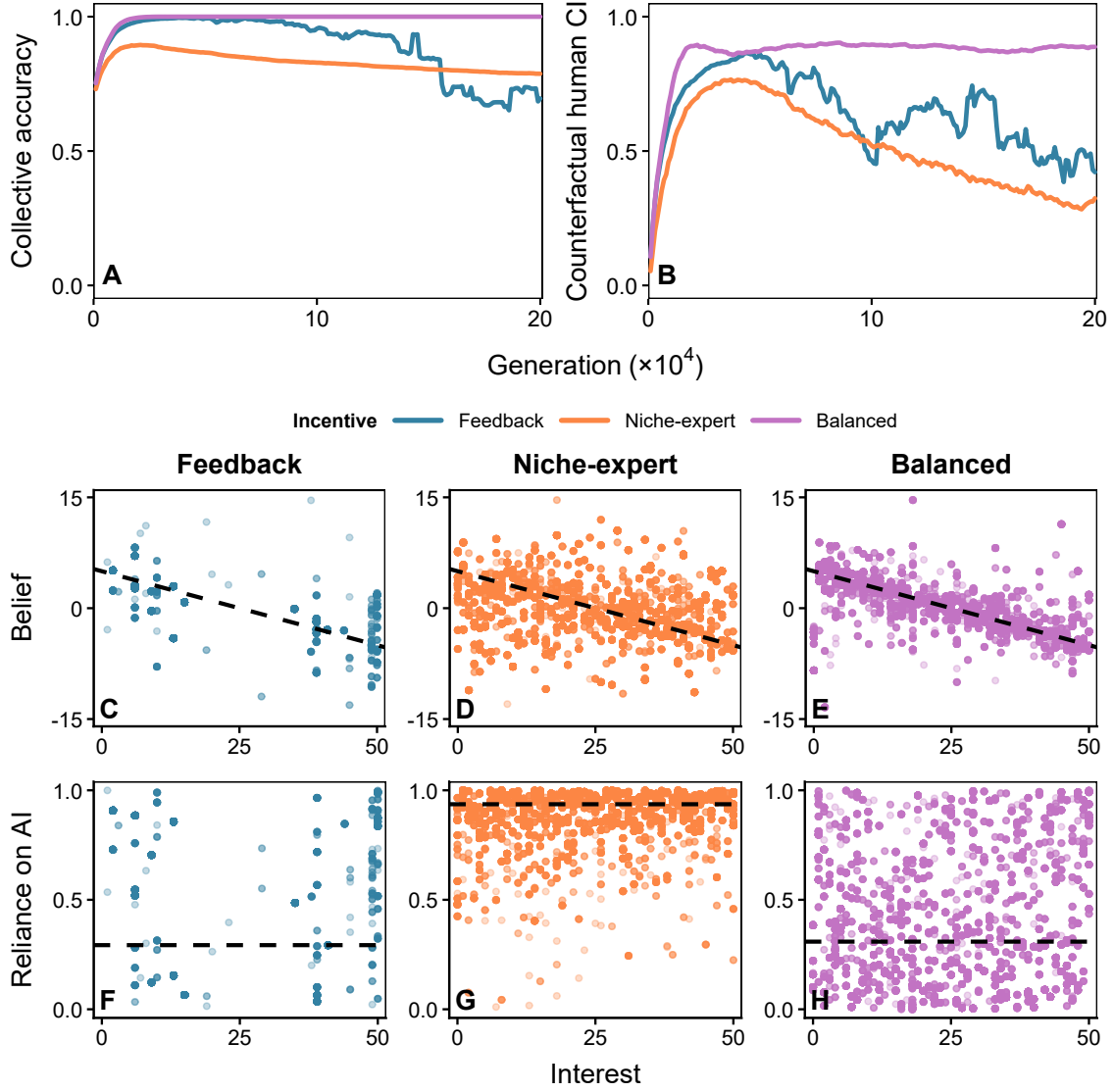


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

Discussion

The ongoing race towards the development of “better” AIs continues to pose questions about the societal benefits and risks of AI uses. Here, we used mathematical models of evolutionary dynamics to investigate how AI uses and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI uses, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research [31] to creative writing [17]. Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions, causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI uses need to be calibrated to prevent systematic AI biases from becoming collective biases.

Our analyses of counterfactual collective accuracy also raises an important concern for future AI uses: while AI uses may improve the apparent performance of individuals and, in some cases, the population as a whole, it can simultaneously reduce the population’s ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assistance, performed worse on their own than those who did not have access [32]. These findings highlight the need to evaluate AI tools not only by the performance gains they provide during use, but also by whether individuals and populations retain the ability to perform without them. Making this distinction clear will be particularly important for education, where AI may help students complete a task without helping them learn how to do it on their own.

As generative AIs and large language models are already generating substantial benefits, many organizations, both industrial and academic, are increasingly seeking to improve their collective performance by regulating AI uses. Such regulations may

266 take direct forms, including AI-use policies or token-based leaderboards, or may op-
267 erate indirectly through other incentives, such as productivity rewards or the social
268 stigma associated with AI use. However, our model suggests that directly incentiviz-
269 ing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
270 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
271 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
272 venting groups from using AI even when it performs well in the relevant domain.
273 Our result instead shows that appropriate AI use and accurate CI can be promoted
274 by focusing on the traditional properties of groups that have been emphasized for
275 decades: maintaining diversity and preventing collective bias, rather than on AI use
276 itself.

277 There are several limitations to our analyses. First, our model allows for so-
278 cial learning among humans, but AI’s inherent ability remains fixed throughout the
279 evolutionary process. In reality, AI systems can also learn from human-generated
280 data. Such bidirectional learning increase the similarity between human and AI er-
281 rors and therefore further reduce the group performance. Second, we model AI errors
282 using fixed, factor-specific additive biases and independent random errors. This as-
283 sumption allows random errors to be reduced through aggregation, while systematic
284 biases can still persist in the collective prediction. In reality, AI errors may have
285 more complex structures, including biased estimates of factor effects and correlated
286 errors across individuals, factors, or repeated interactions. Further empirical and
287 theoretical work is therefore needed to understand how these error structures affect
288 the emergence of CI. Finally, following established models of CI, we assume that
289 the prediction target is a linear combination of independent causal factors. This
290 assumption does not hold for many real-world tasks, where causal factors may be
291 correlated or interact nonlinearly. Future studies should therefore investigate how
292 nonlinear relationships between causal factors and prediction targets affect the in-
293 teraction between AI assistance and the emergence of CI.

294 The rapid adoption of personalized AI assistants is changing how individuals
295 learn, work, and make decisions. While the immediate benefits and risks of these
296 systems have been evaluated at the individual level, our study raises a broader con-
297 cern that their widespread AI use may reshape collective intelligence of human so-
298 ciety. Our results further underline the importance of designing reward structures
299 that preserve minority expertise and prevent AI biases from persisting across the
300 population. **[SWP: need a concluding sentence!]**

301 Methods

302 Prediction target

303 Following previous models of CI, we model the prediction target as a linear combi-
304 nation of M independent causal factors [30, 12, 13]:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

305 where the coefficient α_i determines the magnitude and direction of the effect of
306 factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an
307 independent Gaussian random variable with mean zero and standard deviation σ_i .

308 Players

309 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
310 prediction task. In the absence of AI assistance, the strategy of player A is char-
311 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
312 factor X_{e_A} and makes a prediction by multiplying the observed factor value by their
313 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

314 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does
315 not observe any causal factor and instead makes a constant prediction based solely
316 on their belief. We refer to this as a stubborn strategy because the player’s predic-
317 tion does not respond to the realized values of the causal factors. For notational
318 convenience, we assume that $X_0 = 1$.

319 When the group is assisted by AI, each player’s strategy additionally includes
320 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
321 pendently and revises their prediction by taking a weighted average of their own
322 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

323 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI,
324 whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

325 AI models

326 We assume that the AI model has been trained on similar tasks and is therefore
327 capable of making predictions based on causal factors. However, because the model
328 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
329 The additive bias associated with factor i is denoted by b_i , and the stochastic error
330 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
331 biases therefore remain fixed throughout the simulation. For simplicity, we assume

332 $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows
 333 biases to differ among factors.

334 We assume that the stochastic error follows a Gaussian distribution with mean
 335 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 336 dent error is generated. Errors are therefore independent across players and across
 337 repeated consultations by the same player. Consequently, stochastic AI errors can
 338 be attenuated when independently generated predictions are aggregated across play-
 339 ers. We therefore hold τ constant and vary the magnitude of the additive bias to
 340 manipulate the level of AI accuracy during our analyses.

341 We consider two extreme types of AI that differ in the amount of information
 342 available for prediction. The first is termed the Omniscient AI, which can observe
 343 all M causal factors and incorporate them into its prediction regardless of the user's
 344 interest. The prediction generated by the Omniscient AI for player A is expressed
 345 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

346 The second is termed the Chatbot AI, which can use only the single causal factor
 347 X_{e_A} associated with the player's interest. The prediction generated by the Chatbot
 348 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

349 Aggregation rules

350 The revised individual predictions are aggregated to form a collective prediction. We
 351 use two aggregation rules proposed in previous research [13], depending on the type
 352 of AI used by the group. First, when the group is assisted by the Omniscient AI, the
 353 AI component of each revised prediction contains information about all M factors,
 354 even though the individual's own prediction is based on only one factor. Each revised
 355 prediction can therefore be treated as an estimate of the entire target. Accordingly,
 356 we calculate the collective prediction by taking the simple average of the revised
 357 individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

358 Second, when the group is assisted by the Chatbot AI, both the player's initial
 359 prediction and the AI's prediction are based on the same causal factor. Each revised
 360 prediction therefore contains information about only one factor and can be treated
 361 as an estimate of that factor's contribution to the target. We consequently use
 362 clustering aggregation.

363 Clustering aggregation is performed by grouping individuals according to their
 364 interests and calculating the mean prediction within each group. The collective
 365 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A 's revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

Previous research has shown that these two incentive structures can produce nearly perfect CI through social learning, except when the niche-expert incentive is combined with simple averaging. They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

To examine the effects of directly encouraging or discouraging AI use, we modify the feedback and niche-expert structures by adding a term proportional to the player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

Here, λ is a weighting coefficient representing the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the arithmetic mean of the feedback and niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

Simulations

All simulations were conducted using an agent-based model with 50 causal factors ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction target were sampled at the beginning of the simulations and subsequently held fixed.

The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization. The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

Players' initial strategies were also sampled from predefined probability distributions. The distribution of individual belief c_A depended on the aggregation rule. Under simple averaging, each factor's contribution is scaled by the proportion of players interested in that factor. We therefore sampled beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work [13]. Under clustering aggregation, beliefs were sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player was sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assumed that prediction events occurred on a substantially faster timescale than social learning. At each generation, we therefore calculated the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs were derived for all incentive structures and are presented in the Supplementary Methods.

Each generation consisted of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , were randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduced a mutation rate μ , such that with probability μ , player A did not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate was set to zero in all simulations except those presented in Supplementary Figure 4.

Unless otherwise specified, each simulation was run for 200,000 generations. When a result was represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

434 To evaluate the dynamics of collective prediction, we defined six summary met-
 435 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 436 accuracy, interest diversity, and median reliance on AI. At each population state,
 437 expectations and variances were evaluated analytically over the distributions of the
 438 causal factors and AI errors rather than estimated from sampled prediction events.
 439 Collective bias and collective variance constitute the bias–variance decomposition of
 440 the group level collective prediction error.

441 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
 442 ative value indicates that the collective prediction performs worse than the baseline.
 443 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
 444 the evolved interests and beliefs of the AI-assisted population. In calculating interest
 445 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

446 **Data and code availability**

447 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
448 `parklab-snu/collective-intelligence-with-AI`).

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453 **Contributions**

454 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
455 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
456 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
457 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

458 **Competing Interests**

459 The authors declare no competing interests.

460 Supplementary Text

461 1 Omniscient AI

462 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
463 player's interest.

464 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

465 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

466 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
467 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
468 with zero mean, τ standard deviation with no correlation within or between players.
469 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

471 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
472 β_A stands for reliance on AI.

473 1.2 Aggregation

474 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
475 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \\ &= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \end{aligned}$$

476 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

477

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

478

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

479 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
480 i which is solely based on players' belief.

481 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

482 1.3 Expected payoffs

483 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

484

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

485

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

486 See equation Eq. (S6) for the $\bar{\eta}$.

487 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

488 **Niche-expert**

489 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 490 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

491 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

492 **Incentivizing or penalizing AI use**

493 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 494 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

495 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

496

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

497

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

498 2 Chatbot AI

499 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

500 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 501 mean, τ standard deviation with no correlation within or between players. Player
 502 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

503

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

504 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 505 β_A stands for reliance on AI.

506 2.2 Aggregation

507 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

508 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

509

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

510

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

511

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

512

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

513 2.3 Expected payoffs

514 Feedback

515

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

516

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

517 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

518 Niche-expert

519 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

520 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

521 Incentivizing or penalizing AI use

522 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
523 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

524 Balanced

525 Expected payoff of the balanced structure can also be driven by arithmetic mean of
526 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

527 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

530 3 Evolutionary dynamics

531 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
532 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
533 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

534 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
535 with positive frequency have the same expected payoff. A boundary state can also
536 be stationary because a strategy with exactly zero frequency cannot increase under
537 the replicator equation. Thus, stability of a boundary state needs to be considered
538 separately by examining invasion by a rare strategy.

539 3.1 Omniscient AI

540 3.1.1 Feedback

541 When the population size approaches infinity, the effect of stochastic error on the
542 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
543 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
544 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

545 The function V is non-negative and equals zero when the collective prediction is
546 identical to the target. To examine how V changes under the evolutionary dynamics,
547 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

548 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

549 Although K changes with the population state, it is common to all strategies at a
550 given state and can therefore be treated as a constant with respect to the strategy
551 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

552 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

553 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

554 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

555 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
556 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

557 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

591 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 592 the expected squared prediction error at the corresponding boundary.

593 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

594 Substituting this into the expected payoff gives a strictly concave function of β ,
 595 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

596 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

597 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

598 We denote the optimal strategy, or the payoff-equivalent strategy class when
 599 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 600 be stationary if the optimal strategy is absent, but it is unstable against invasion
 601 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 602 a stable state can contain only S_e within each represented interest.

603 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 604 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

605 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 606 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

607 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

608 This internal stationary state is locally stable. At equilibrium, all represented
609 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
610 interest increases slightly above its equilibrium value, its payoff decreases because
611 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
612 an interest decreases below its equilibrium value, its payoff increases. The replicator
613 dynamics therefore generate a restoring force that reduces small deviations in interest
614 frequencies and drives the population back toward the internal stationary state.

615 If one or more interests are absent, the resulting boundary state can be stationary
616 under the replicator equation, but it is unstable against invasion. The payoff of a rare
617 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
618 population is negative when $L_e > 0$, so the rare interest increases when introduced
619 at sufficiently small frequency.

620 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
621 prediction strategies and their frequencies. Whether the constrained optimum has
622 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
623 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
624 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
625 interest frequency remain uniquely determined.

626 This equilibrium also allows a direct comparison with the expected accuracy of
627 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

628 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
629 fore, because L_e is the minimum individual expected squared prediction error within
630 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

631 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

632 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

633 The collective prediction is the average of individual revised predictions. By
634 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

Because the collective accuracy and AI accuracy use the same normalization by $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is strictly greater than the expected accuracy of the AI prediction.

Although the niche-expert structure improves collective prediction relative to the AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$.

3.2 Chatbot AI

3.2.1 Feedback

With Chatbot AI, the effect of stochastic error on the collective prediction depends on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the stochastic error to vanish. However, Under this assumption, the loss of interest diversity cannot occur, and therefore the stochastic breakdown of collective accuracy observed in finite populations cannot be analytically captured. Instead, by assuming that each interest is represented by infinitely many players, we can show that the expected collective prediction error monotonically decreases until perfect accuracy is achieved. Because finite population dynamics are more relevant to the empirical populations considered in this study, we retain finite population simulations for the main result.

3.2.2 Niche-expert

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

664 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 665 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

666 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 667 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 668 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 669 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 670 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 671 the emergence of CI. Only when the prediction task is centered near zero and AI is
 672 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 673 exist.

674 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

675 Substituting this into the expected payoff gives a strictly concave function of β , with
 676 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

677 **Supplementary Figure**

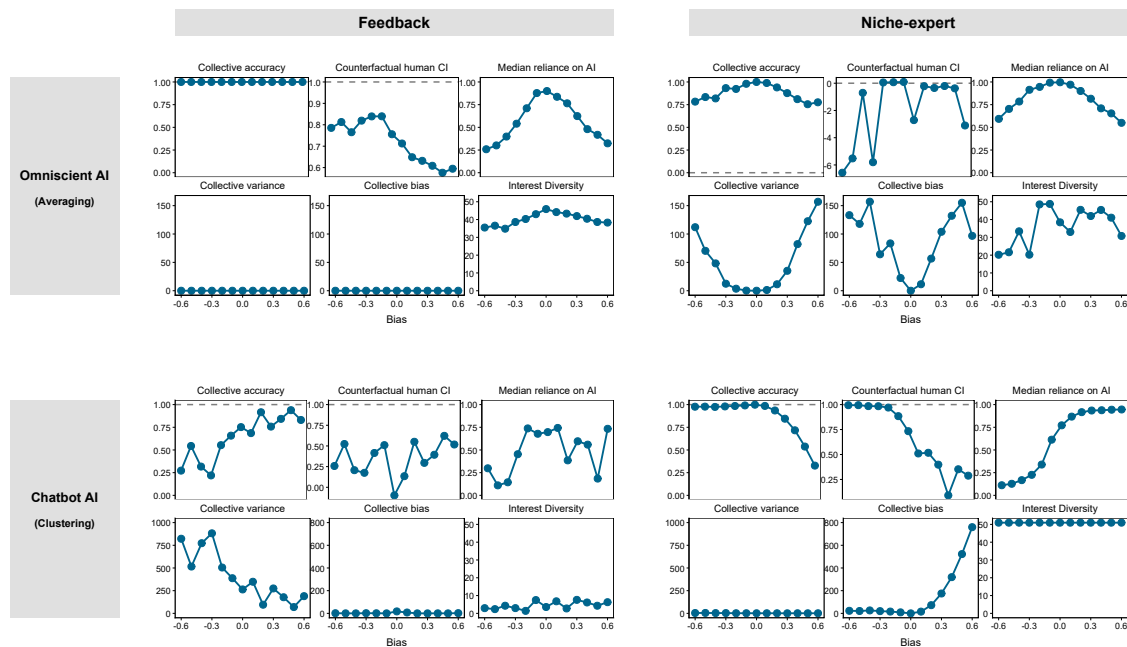


Figure S1: Caption. Text.

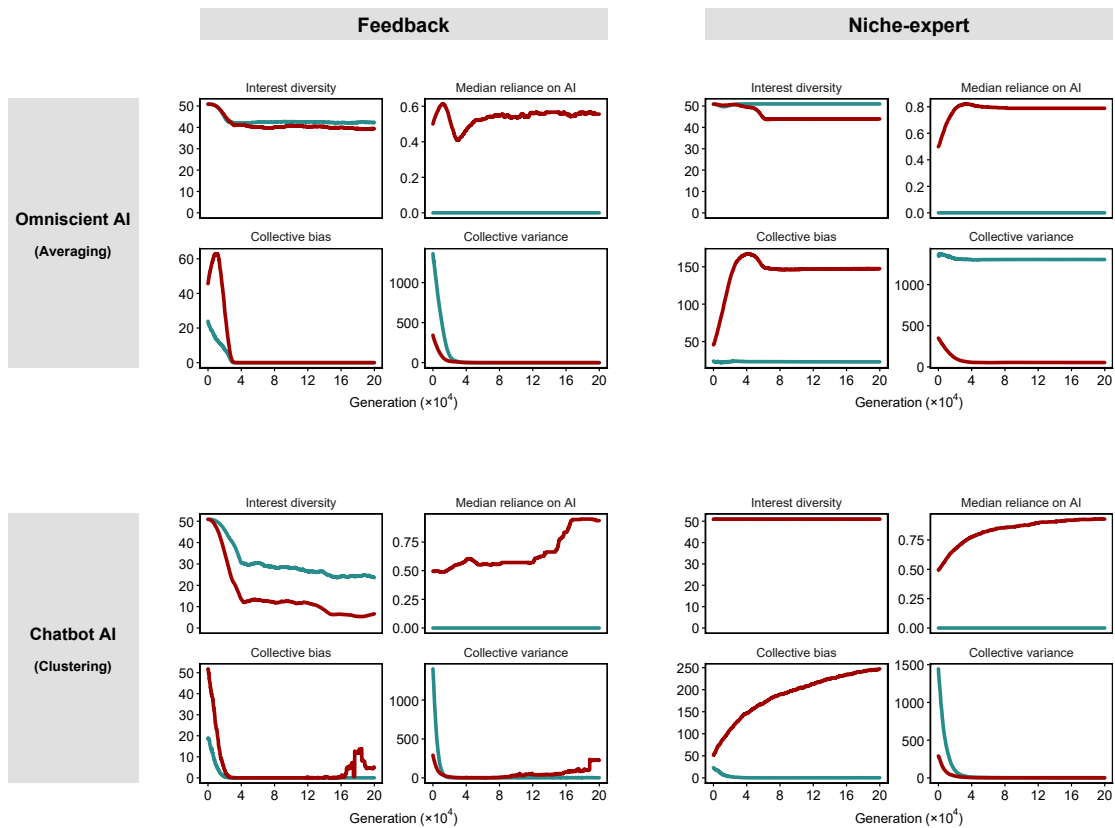


Figure S2: **Caption.** Text.

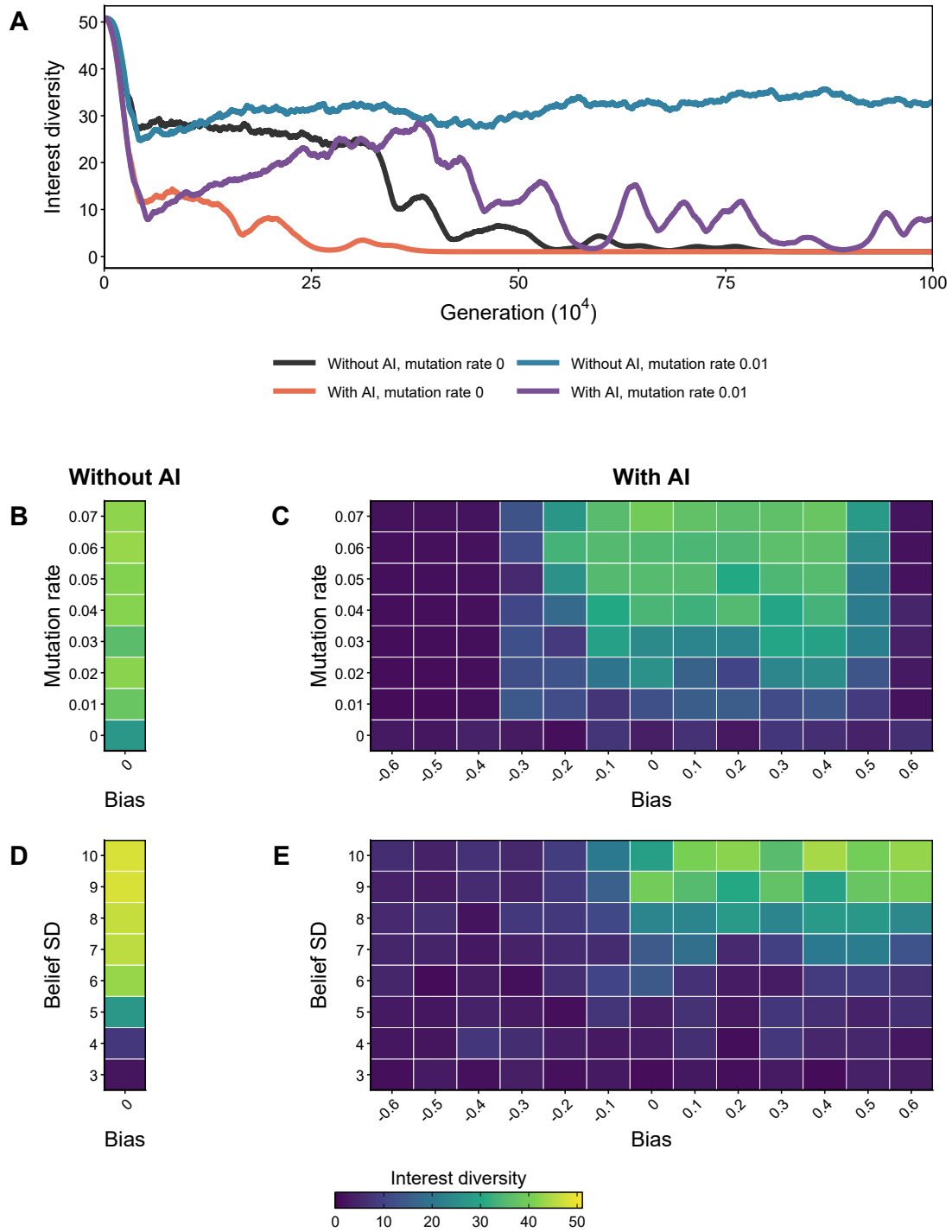


Figure S3: **Caption.** Text.

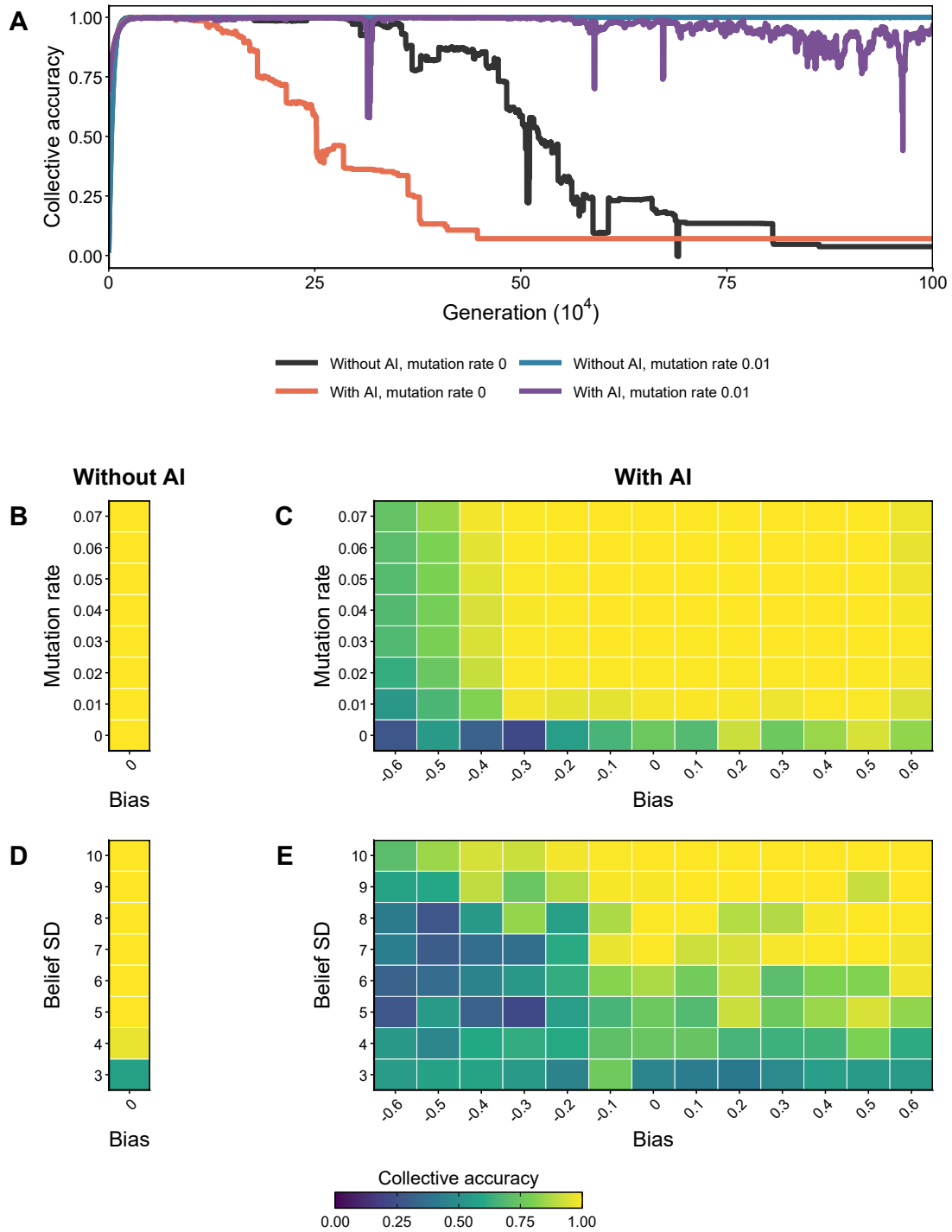


Figure S4: **Caption.** Text.

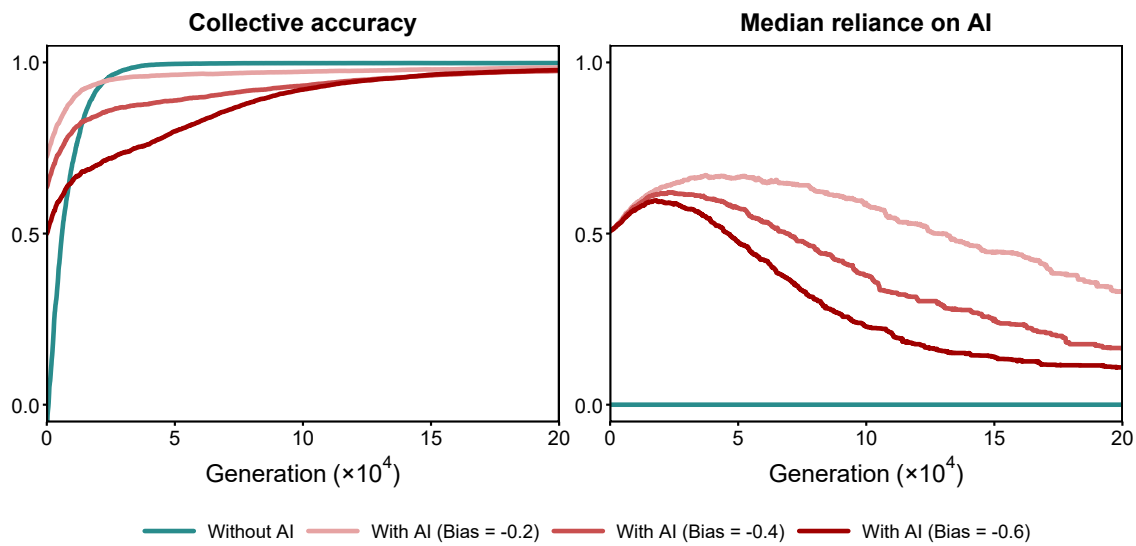


Figure S5: **Caption.** Text.

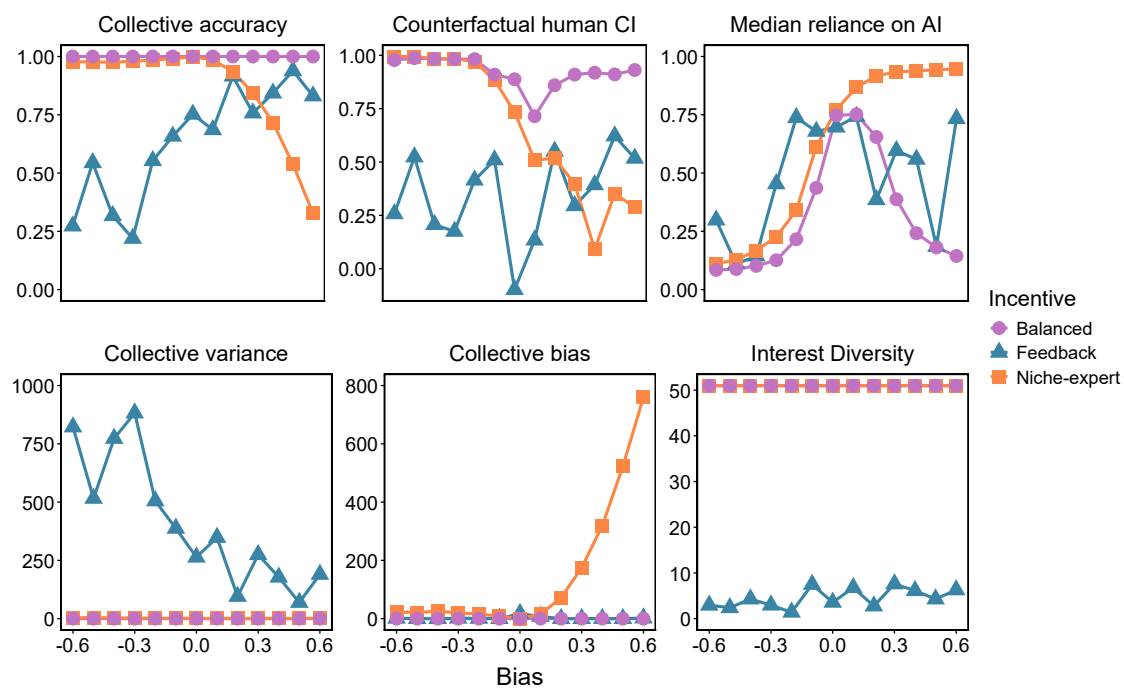


Figure S6: **Caption.** Text.

References

- [1] Francis Galton. Vox populi, 1907.
- [2] Anita Williams Woolley, Christopher F Chabris, Alex Pentland, Nada Hashmi, and Thomas W Malone. Evidence for a collective intelligence factor in the performance of human groups. *science*, 330(6004):686–688, 2010.
- [3] Ilan Yaniv and Maxim Milyavsky. Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes*, 103(1):104–120, 2007.
- [4] Cars Hommes, Joep Sonnemans, Jan Tuinstra, and Henk Van de Velden. Coordination of expectations in asset pricing experiments. *The Review of Financial Studies*, 18(3):955–980, 2005.
- [5] Irving Lorge, David Fox, Joel Davitz, and Marlin Brenner. A survey of studies contrasting the quality of group performance and individual performance, 1920–1957. *Psychological bulletin*, 55(6):337, 1958.
- [6] Tabea Dreyer, Amir Haluts, Amos Korman, Nir Gov, Ehud Fonio, and Ofer Feinerman. Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, 122(1):e2414274121, 2025.
- [7] Andrew M Simons. Many wrongs: the advantage of group navigation. *Trends in ecology & evolution*, 19(9):453–455, 2004.
- [8] Andrew Berdahl, Colin J Torney, Christos C Ioannou, Jolyon J Faria, and Iain D Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013.
- [9] Christoph Riedl, Young Ji Kim, Pranav Gupta, Thomas W Malone, and Anita Williams Woolley. Quantifying collective intelligence in human groups. *Proceedings of the National Academy of Sciences*, 118(21):e2005737118, 2021.
- [10] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [11] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [12] Richard P Mann and Dirk Helbing. Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences*, 114(20):5077–5082, 2017.
- [13] Guocheng Wang, Qi Su, Long Wang, and Joshua B Plotkin. Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences*, 122(51):e2516535122, 2025.

- [14] Justin Werfel. Fluid thinking about collective intelligence. *Nature Machine Intelligence*, 8(4):506–516, 2026.
- [15] Hao Cui and Taha Yasseri. AI-enhanced collective intelligence. *Patterns*, 5(11):101074, 2024.
- [16] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651(8107):914–919, 2026.
- [17] Anil R Doshi and Oliver P Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.
- [18] Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarasci, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, et al. Artificial intelligence index report 2026. *arXiv preprint arXiv:2606.15708*, 2026.
- [19] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12):2293–2303, 2024.
- [20] Jason W Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A Bakker, Joshua A Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, et al. How large language models can reshape collective intelligence. *Nature human behaviour*, 8(9):1643–1655, 2024.
- [21] Lennart Meincke, Gideon Nave, and Christian Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature human behaviour*, 9(6):1107–1109, 2025.
- [22] Andreas Fügener, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. Will humans-in-the-loop become borgs? merits and pitfalls of working with AI. *MIS quarterly*, 45(3):1527–1556, 2021.
- [23] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- [24] Lucy M Aplin, Damien R Farine, Richard P Mann, and Ben C Sheldon. Individual-level personality influences social foraging and collective behaviour in wild birds. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789):20141016, 2014.
- [25] Lu Hong and Scott E Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.

- 747 [26] Isabelle Bousquette. Why some companies say AI ‘token-
748 maxxng’ is key to survival. [https://www.wsj.com/cio-journal/
749 why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128](https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128),
750 April 2026. The Wall Street Journal. Accessed: 2026-07-07.
- 751 [27] Tim Keary. Is the cult of ‘tokenmaxxing’ just another fad or the
752 new normal? [https://www.forbes.com/sites/timkeary/2026/04/13/
753 is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/](https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/),
754 April 2026. Forbes. Accessed: 2026-07-07.
- 755 [28] Dalmeet Singh Chawla. Researchers who use hallucinated references to face
756 arXiv ban. *Nature*, 653(8116):988–989, 2026.
- 757 [29] H Holden Thorp. Chatgpt is fun, but not an author. *Science*, 379(6630):313–313,
758 2023.
- 759 [30] Lu Hong, Scott E Page, and Maria Riolo. Incentives, information, and emergent
760 collective accuracy. *Managerial and Decision Economics*, 33(5-6):323–334, 2012.
- 761 [31] Qianyu Hao, Fengli Xu, Yong Li, and James Evans. Artificial intelligence tools
762 expand scientists’ impact but contract science’s focus. *Nature*, 649(8099):1237–
763 1243, 2026.
- 764 [32] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and
765 Rei Mariman. Generative AI without guardrails can harm learning: Evidence
766 from high school mathematics. *Proceedings of the National Academy of Sciences*,
767 122(26):e2422633122, 2025.